

SAYAN CHAKRABARTY

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RESEARCH INTERESTS

Scalable inference for large networks and complex data, interpretability for network-assisted blackbox machine learning models, statistical methods for complex data using randomization, and applications of machine learning in biological sciences and medicine.

EDUCATION

University of Illinois Urbana-Champaign <i>PhD in Statistics</i> Advisors: Yuguo Chen and Srijan Sengupta Collaborators: Fatima Husain and Raksha Mudar Thesis: Subsampling Based Inference for Network Data	Champaign, IL 2024
Indian Statistical Institute <i>MSTAT with Specialization in Computational Statistics Track</i>	Kolkata, India 2018
St. Xavier's College <i>BSc (Honors) in Statistics with Minors in Mathematics and Computer Science</i>	Kolkata, India 2016

POSITIONS

University of Michigan <i>Postdoctoral Research Fellow in Statistics</i> Advisors: Elizaveta Levina and Ji Zhu Collaborators: Snigdha Panigrahi, Shelly Flagel and Akbar Waljee	Ann Arbor, MI 2024 – Present
University of Illinois Urbana-Champaign <i>Graduate Assistant</i>	Champaign, IL 2018 – 2024
Illinois Statistics Office <i>Statistical Consultant</i> Provided statistical consultation, designed experiments, and performed statistical analysis for clients from both academia and industry. Collaborated on research projects in medicine, biology, psychology, geology, finance, law, linguistics, engineering, and political science.	Champaign, IL 2020 – 2024
Synchrony Financial <i>Summer Intern, Information Security and Data Protection Team</i> Analyzed and predicted the number of data breaches and compared the efficacies of multiple data protection measures.	Champaign, IL (Remote) Summer 2021
CHEGG INC. <i>Expert Answerer in Statistics and Probability (Part time)</i>	Kolkata, India 2014 – 2016

PUBLICATIONS

Accepted / Under Review

Chakrabarty, S., Sengupta, S., Chen, Y. (2026+). Network Cross-Validation and Model Selection via Subsampling. *Minor revision at Journal of the Royal Statistical Society Series B*, arXiv preprint:2504.06903. [Preprint](#).

Chakrabarty, S., Sengupta, S., Chen, Y. (2025). Subsampling Based Community Detection for Large Networks. *Statistica Sinica* (35) 1627 – 1648. [Article](#).

Moran, C., Park, T., Buffenstein, R., **Chakrabarty, S.**, Lindeblad, M., Fortman, J., Adams, C. (2024) Pharmacokinetics of Injectable Meloxicam and Buprenorphine in the Naked Mole-Rat (*Heterocephalus glaber*). *Journal of the American Association for Laboratory Animal Science* (63) 565 – 571. [Article](#)

Chakrabarty, S., Mudar, R., Chen, Y., Husain, F. (2024). Contribution of Tinnitus and Hearing Loss to Depression: NHANES Population Study. *Ear and Hearing* (45) 775 – 786. [Article](#)

Han, M. D., Kwon, T. G., Miloro, M., **Chakrabarty, S.** (2023). What Is the Linear Accuracy of Regional Voxel-Based Registration for Orthognathic Surgery Landmarks?. *Journal of Oral and Maxillofacial Surgery* (81) 546 – 556. [Article](#)

Ghosh, A., SahaRay, R., **Chakrabarty, S.**, Bhadra, S. (2021). Robust Generalised Quadratic Discriminant Analysis. *Pattern Recognition* (117) 107981. [Article](#)

In Preparation

Chakrabarty, S., Levina, E. Resampling Networks and Network Statistics.

Chakrabarty, S., Levina, E. Fast Network Subsampling and Aggregation Using Overlapping Partitions.

Chakrabarty, S., Panigrahi, S., Levina, E. Randomized Network Cross-Validation.

Chakrabarty, S., Levina, E., Zhu, J. Local Feature Importance in Network Linked Machine Learning Prediction Models.

Ramos, B., **Chakrabarty, S.**, Cain, C., Levina, E., Kennedy, R., Flagel, S. Identifying a Neurochemical Signature of the Paraventricular Nucleus of the Thalamus that Predicts Individual Variability in Cue-Reward Learning.

Hou, J., **Chakrabarty, S.**, Huang, J., Van, T., Baker, J., Zhu, J., Waljee, A. Machine Learning Models to Predict Objective Remission in Veterans with Ulcerative Colitis Treated with Thiopurines.

PRESENTATIONS

The 9th International Conference on Econometrics and Statistics (EcoSta) Tokyo, Japan
Invited Talk: Bootstrapping Network Statistics Using Overlapping Partitions 2026

International Symposium on Nonparametric Statistics Thessaloniki, Greece
Invited Talk: Network Bootstrap Using Overlapping Partitions 2026

Recent Advances in Random Networks Chicago, USA
Lightning Talk: Network Bootstrap Using Overlapping Partitions 2026
Poster: Network Cross-Validation and Model Selection via Subsampling

International Conference on Statistics and Data Science Seville, Spain
Contributed Talk: Network Bootstrap Using Overlapping Partitions 2025

International Conference on Computational and Mathematical Statistics London, UK
Invited Talk: Network Bootstrap Using Overlapping Partitions 2025

International New Researchers' Conference Seville, Spain
Poster: Network Cross-Validation and Model Selection via Subsampling 2025

Joint Statistical Meetings Nashville, USA
Contributed Talk: Network Bootstrap Using Overlapping Partitions 2025

International Indian Statistical Association Conference Lincoln, USA
Invited Talk: Network Bootstrap Using Overlapping Partitions 2025

Department of Statistics, North Carolina State University Raleigh, USA
Course Presentation: Network Cross-Validation and Model Selection via Subsampling 2025

The Michigan Student Symposium for Interdisciplinary Statistical Sciences Ann Arbor, USA
Contributed Talk: Network Bootstrap Using Overlapping Partitions 2025

Symposium on New Frontiers in Reliability and Risk Analysis Washington, D.C., USA
Invited Poster: Subsampling Based Model Selection for Network Data 2023

Joint Statistical Meetings Toronto, Canada
Contributed Talk: Subsampling Based Model Selection for Network Data 2023

International Indian Statistical Association Conference Golden, USA
Student Poster: Subsampling Based Community Detection for Large Networks 2023

Joint Statistical Meetings Washington, D.C., USA
Invited Talk and Poster: Subsampling Based Community Detection for Large Networks 2022

Institute of Mathematical Statistics Annual Meeting London, UK
Contributed Talk: Subsampling Based Community Detection for Large Networks 2022

New England Statistics Symposium Storrs, USA
Invited Student Talk: Subsampling Based Community Detection for Large Networks 2022

Bohrer Workshop Champaign, USA
Student Talk and Poster: Subsampling Based Community Detection for Large Networks 2022

AWARDS AND ACHIEVEMENTS

Travel Award, <i>Recent Advances in Random Networks</i> , Institute of Mathematical and Statistical Innovation	2026
Travel Award, <i>International New Researchers' Conference</i> , Institute of Mathematical Statistics	2025
Travel Award, <i>International Indian Statistical Association Conference</i>	2025
Travel Award, <i>Symposium on New Frontiers in Reliability and Risk Analysis: A Tribute to Nozer D. Singpurwalla</i> , George Washington University	2023
Awarded 'Excellent in Teaching', <i>University of Illinois Urbana-Champaign</i>	2019
All India Rank 8 in Mathematical Statistics at <i>Indian Institute of Technology Joint Admission Test for Masters</i>	2016
INSPIRE SHE Fellowship, <i>Department of Science & Technology, Govt. of India</i>	2013 – 2018

TEACHING EXPERIENCE

Department of Statistics, University of Illinois Urbana-Champaign	Champaign, IL
<i>Instructor</i>	
STAT 200: Introduction to Statistical Analysis	Fall 2019
<i>Discussion Leader</i>	
STAT 400: Statistics and Probability I	Fall 2018, Spring and Summer 2019
<i>Teaching Assistant</i>	
STAT427: Statistical Consulting	Spring 2021, Spring 2022, Spring 2023
STAT578: Doctoral Topic Course - Quantile Regression	Fall 2022
STAT511: Advanced Mathematical Statistics	Fall 2021
STAT430: Graduate Topic course - Deep Learning	Fall 2020
STAT578: Doctoral Topic course - Advanced Bayesian Analysis	Spring 2020
Department of Mathematics, University of Illinois Urbana-Champaign	Champaign, IL
<i>Workshop Co-Host and Student Project Mentor</i>	
PI4 Computation and Data Science Boot Camp	Summer 2019

PROFESSIONAL SERVICE

Journal Reviewer	2024 – Present
<i>Journal of the American Statistical Association</i>	
<i>Computational Statistics and Data Analysis</i>	
<i>Journal of Computational and Graphical Statistics</i>	
<i>Journal of the Royal Statistical Society: Series B</i>	
University Service	
<i>Postdoctoral Representative to the Department of Statistics Cultural Committee</i> , University of Michigan	2025 – 2026
<i>Founding Chair of the Doctoral Student Seminar Series at the Department of Statistics</i> , University of Illinois Urbana-Champaign	2022
<i>Member of the Statistics Doctoral Student Association at the Department of Statistics</i> , University of Illinois Urbana-Champaign	2021 – 2023
<i>Founding Member of the PhD Student Seminar Series at the Department of Statistics</i> , University of Illinois Urbana-Champaign	2019 – 2020
Volunteer Work	
<i>Membership Committee</i> , International Indian Statistical Association	2025 – Present

SOFTWARE

netOP: R package for network operations using overlapping partitions.

COMPUTING SKILLS

Programming Languages: R, Rcpp, Python, SPSS, SAS, C/C++

Miscellaneous: Git/GitHub, Linux, HPC, Docker/Singularity